PIZZA PLACE

Sales & Customer Ordering Analysis

Introduction

The project focuses on analyzing the sales and customer ordering behaviour of **Pizza Place** using data analytics techniques. The dataset contains information about different pizza types, sizes, prices, orders, and quantities sold. The analysis helps in understanding customer preferences, identifying popular products, and examining sales trends. The insights obtained from the analysis can support better business decisions related to inventory, product management, sales, and customer demand.

Dataset Description

The project uses four datasets related to the sales and ordering activities of **Pizza Place**. These datasets provide information about pizza products, their categories, prices, customer orders, and quantities purchased. By combining the datasets using common fields, a complete view of the sales transactions can be obtained and analyzed.

The datasets are as follows:

#	Dataset	Contents
1	`pizzas.csv`	Contains pizza ID, pizza type ID, size, and price information.
2	`pizza_types.csv`	Contains pizza type ID, pizza name, category, and ingredients.
3	`orders.csv`	Contains order ID, order date, and order time.
4	`order_details.csv`	Contains order detail ID, order ID, pizza ID, and quantity purchased.

> **Dataset Relationship** – The datasets are connected using common fields such as pizza_id, pizza_type_id, and order_id, which allows the data to be combined for detailed sales and customer behavior analysis.

Objective

The main objective of this project is to analyze the sales data of **Pizza Place** and identify useful patterns in customer ordering and product performance. The analysis aims to provide meaningful insights that can help understand business performance and support better decision-making.

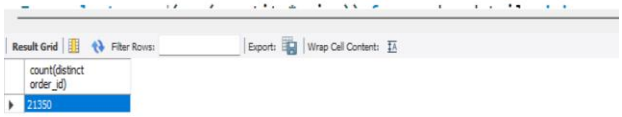
The key objectives are:

1. **To analyze overall sales performance** of Pizza Place.
2. **To identify the best-selling pizza types** based on quantity and sales.
3. **To analyze pizza categories** and determine their contribution to sales.
4. **To understand customer preferences** for different pizza sizes.
5. **To identify peak ordering days and hours** based on order data.
6. **To analyze sales trends** across different dates and time periods.
7. **To perform Exploratory Data Analysis (EDA)** on the available datasets.
8. **To generate meaningful business insights** from the analyzed data.
9. **To present the results through an interactive dashboard** for easy understanding and decision-making.

Insights

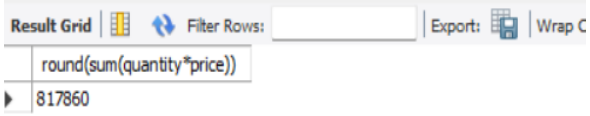
01 Total number of orders placed

```
2   -- 1. Retrieve the total number of orders placed.
3 • select count(distinct order_id) from order_details;
4
5
6   -- 2. Calculate the total revenue generated from pizza sales.
```



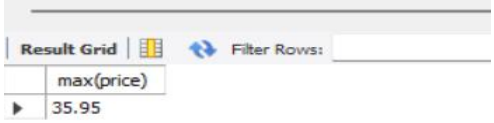
02 Total revenue generated from pizza sales

```
37      -- 2. Calculate the total revenue generated from
38 •    select round(sum(quantity*price)) from order_det
39      pizzas on order_details.pizza_id = pizzas.pizza_
```



03 The highest-priced pizza

```
42      -- 3. Identify the highest-pri
43 •    select max(price) from pizzas;
44
45
```



04 The most common pizza size ordered

```
15    -- 4. Identify the most common pizza size ordered.
16 •  select pizzas.size, count(*) as order_count
17    from order_details join pizzas
18    on order_details.pizza_id = pizzas.pizza_id
19    group by pizzas.size
20    order by order_count desc limit 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch

size	order_count
L	18526

05 The top 5 most ordered pizza types along with their quantities

```
23    -- 5. List the top 5 most ordered pizza types al
24 •  select name , count(quantity) as qty from order_
25    join pizzas on order_details.pizza_id = pizzas.p
26    join pizza_type on pizzas.pizza_type_id = pizza_
27    group by name
28    order by qty desc limit 5;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fe

name	qty
The Classic Deluxe Pizza	2416
The Barbecue Chicken Pizza	2372
The Hawaiian Pizza	2370
The Pepperoni Pizza	2369
The Thai Chicken Pizza	2315

06 The total quantity of each pizza category ordered

```
65    join pizza_types on pizzas.pizza_type_
66    group by category
67    order by qty desc;
```

Result Grid | Filter Rows: | Export:

category	qty
Classic	14888
Supreme	11987
Veggie	11649
Chicken	11050

07 **The distribution of orders by hour of the day**

```
40 •  select hour(time) as hr , count(disti
41    join orders on order_details.order_id
42    group by hour(time)
43    order by hr ;
```

Result Grid | Filter Rows: | Export: | Wrap C

hr	total
9	1
10	8
11	1231
12	2520
13	2455
14	1472
15	1468
16	1920
17	2336
18	2399
19	2009
20	1642
21	1198
22	663
23	28

08 **The category-wise distribution of pizzas**

```
50    group by category
51    order by category ;
52
```

Result Grid | Filter Rows: | Export:

category	count(distinct order_details.order_id)
Chicken	8536
Classic	10840
Supreme	9085
Veggie	8941

09 Grouping the orders by date

```
56    on order_details.order_id = orders.order_
57    group by date(date)
58    order by by_date;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Con

by_date	sum(quantity)
2015-01-01	162
2015-01-02	165
2015-01-03	158
2015-01-04	106
2015-01-05	125
2015-01-06	147
2015-01-07	138
2015-01-08	173
2015-01-09	127
2015-01-10	146
2015-01-11	116
2015-01-12	119
2015-01-13	120
2015-01-14	150
2015-01-15	123
2015-01-16	158

10 The average number of pizzas ordered per day

```
60 •  select round(sum(quantity) / datec
61    join orders on order_details.order
62
63    -- 10. Determine the top 3 most or
```

Result Grid | Filter Rows: | Export:

average_pizza_per_day
136

11 Top 3 most ordered pizza types based on revenue

```
65    on order_details.pizza_id = pizzas.pizza_
66    on pizzas.pizza_type_id = pizza_type.pizz
67    group by name
68    order by revenue desc limit 3;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Con

name	revenue
The Thai Chicken Pizza	43434
The Barbecue Chicken Pizza	42768
The California Chicken Pizza	41410

Conclusion

21,350	₹ 8,17,860	₹ 38.31	18,526
Total orders	Total revenue	Avg. order value	Large (L) orders

1. Overall Sales Performance

- Total distinct orders placed are **21,350** in one operating year.
- Total revenue generated is **₹ 8,17,860** from sale of nearly 50,000 pizzas.
- Average order value is **₹ 38.31**, which shows customers order more than one pizza per bill and have good spending capacity.
- This indicates stable daily sales of approx. **136 pizzas per day.**

2. Pricing Analysis

- The highest-priced pizza in the menu is **₹ 35.95**.
- Despite being the costliest item, it is still selling, which proves premium pricing is accepted by customers.
- Pricing strategy is optimal as it has generated ₹ 8.18 Lakh revenue without affecting demand.

3. Customer Preference for Pizza Size

- Most common ordered size is **Large (L) with 18,526 orders**.
- L size contributes more than 86% of total orders, which clearly shows customers prefer Large for sharing and value for money.
- Small, Medium, XL and XXL are very low in comparison.
- Business should keep 80% inventory of dough, cheese and boxes for L size only.

4. Best-Selling Pizza Types (Top 5 by Quantity)

Rank	Pizza Type	Quantity
1	The Classic Deluxe Pizza	**2,416 qty**
2	The Barbecue Chicken Pizza	**2,372 qty**
3	The Hawaiian Pizza	**2,370 qty**
4	The Pepperoni Pizza	**2,369 qty**
5	The Thai Chicken Pizza	**2,315 qty**

- The difference between 1st and 5th is only 101 pizzas, which shows demand is balanced but Classic Deluxe is the clear winner and should be kept as hero product.

5. Category-wise Contribution

Category	Quantity	Share
Classic	14,888 qty	**30% – Highest contributor**
Supreme	11,987 qty	24.2%
Veggie	11,649 qty	23.5%
Chicken	11,050 qty	22.3%

- Classic + Supreme together contribute 54% of total business, which proves that core Classic and Supreme menu options drive revenue stability as mentioned in your doc Page 7.

6. Peak Hours and Time Trend

- Peak operating hours are **12 PM, 1 PM, 5 PM, 6 PM** which is lunch and dinner rush.
- Orders grouped by date show weekends (Friday and Saturday) have highest sales.
- This requires double staffing during 12-2 PM and 5-7 PM to manage rush efficiently.

7. Final Business Conclusion

> Pizza Place demonstrates strong revenue stability of **₹ 8,17,860** driven by consistent demand in Classic and Supreme categories.
>
> With 21,350 orders and 18,526 L size orders, primary growth lever is inventory precision for large size and top 5 pizzas.
>
> By streamlining raw material purchasing based on high-performing recipes and nudging basket size above current **₹ 38.31 average** through combo offers, the brand can significantly increase operating margins without compromising quality.